

Dmitry Glukhov

AI / Software Engineer

Coimbra, Portugal | Europe/Lisbon | dimkin3@gmail.com | [GitHub](#) | [Telegram](#)

Profile

AI / Software Engineer building evidence-driven AI, data and research products. I turn complex source material, model outputs and noisy datasets into inspectable systems with explicit provenance, durable state, evaluation, failure boundaries and useful interfaces. My recent work spans temporal evidence systems, LLM memory experiments, historical media reconstruction, executable generative graphics, OCR, speech-to-text and scientific pipelines.

Selected Projects

Reality Wiki - temporal evidence and model memory | Private R&D

- Built an append-only epistemic ledger that separates publication, availability, ingestion and event time, with historical and operational as-of replay and SHA-256 evidence hashes.
- Mirrored and normalized official CBR and Rosstat sources into cutoff-safe SQLite, DuckDB and Parquet layers with explicit provenance and fail-closed source boundaries.
- Implemented a 26-origin walk-forward memory study with 104 resolved questions per arm and a stricter forward-only experiment. Retrospective results are explicitly not presented as forecasting skill.

Time Receiver - historical media PWA | [Live product](#) | [Public release](#)

- Built a static-first radio and television environment for Moscow across 18,628 days from 1940 to 1990.
- Packaged 4.09M radio and 395K TV events into 51 on-demand yearly bundles with an offline-capable PWA shell.
- Designed source-aware fidelity labels for exact schedules, reconstructed gaps and equivalent media, plus archival ingestion and editorial-review pipelines.

Form / Field - executable generative graphics laboratory | [Live lab](#) | [Source](#)

- Built a Vue/Vite/p5.js research interface around 34 clearly attributed sketches and a code snapshot of 845 community works by 26 authors.
- Made the visible RAW code the actual rendering source; added bounded genome compilation, manual 3D projection, topology tools and stateful entities.
- Shipped responsive lab, archive, evolution and theory surfaces with 16 Node test modules and automated Pages deployment.

Native Audio Translator | [Repository](#)

- Built a native macOS prototype for live audio translation and OCR subtitle overlays with separate capture, recognition, stabilization, translation and rendering stages.
- Added runtime permission flows, command control, validation executables, local diagnostics and app bundling.

Additional Public Work

- [Feynman Reader](#) - bilingual static reader for 3 volumes and 115 chapters, with 12,644 formulas, 1,038 figures, media manifests and validation tooling.
- [AI Agent Case Studies](#) - restartable Telegram-to-wiki LLM pipeline with state, retries, token visibility and operational diagnostics.
- [Mermaid Compiler](#) - AI-assisted diagram workspace with validation, repair loops and multi-provider routing.

Skills

TypeScript, Python, Swift, Rust, Vue, React, Vite, SQLite, Postgres, DuckDB, Parquet, Docker, GitHub Actions, LLM pipelines, temporal replay, model evaluation, OCR, speech-to-text, data provenance, PWA, Web Audio, scientific computing and visual QA.

Education

Candidate of Sciences in Chemistry (PhD equivalent)

Kazan National Research Technological University